```
!unzip -o /content/gpu_assignment2_genai_colab.zip -d /content
%cd /content/gpu_assignment2_genai
!ls
```

```
Archive:  /content/gpu_assignment2_genai_colab.zip
  inflating: /content/gpu_assignment2_genai/.gitignore
  inflating: /content/gpu_assignment2_genai/AI_USE.md
   creating: /content/gpu_assignment2_genai/checkpoints/
   creating: /content/gpu_assignment2_genai/figures/
  inflating: /content/gpu_assignment2_genai/GPU_HOURS.md
  inflating: /content/gpu_assignment2_genai/HW4_5.ipynb
  inflating: /content/gpu_assignment2_genai/METRICS.md
  inflating: /content/gpu_assignment2_genai/README.md
  inflating: /content/gpu_assignment2_genai/requirements.txt
  inflating: /content/gpu_assignment2_genai/RESERVATION.md
   creating: /content/gpu_assignment2_genai/results/
  inflating: /content/gpu_assignment2_genai/RUN_LOG.txt
   creating: /content/gpu_assignment2_genai/system/
  inflating: /content/gpu_assignment2_genai/WRITTEN.md
  inflating: /content/gpu_assignment2_genai/scripts/00_capture_system.py
  inflating: /content/gpu_assignment2_genai/scripts/01_bert_finetune.py
  inflating: /content/gpu_assignment2_genai/scripts/02_custom_inference.py
  inflating: /content/gpu_assignment2_genai/scripts/03_vllm_benchmark.py
  inflating: /content/gpu_assignment2_genai/scripts/04_part_a_batch_scaling.py
  inflating: /content/gpu_assignment2_genai/scripts/05_part_b_memory_techniques.py
  inflating: /content/gpu_assignment2_genai/scripts/06_part_c_ddp.py
  inflating: /content/gpu_assignment2_genai/scripts/07_part_d_serving_and_kv.py
  inflating: /content/gpu_assignment2_genai/scripts/08_make_combined_figures.py
  inflating: /content/gpu_assignment2_genai/scripts/09_write_metrics.py
  inflating: /content/gpu_assignment2_genai/src/bert_engine.py
  inflating: /content/gpu_assignment2_genai/src/config.py
  inflating: /content/gpu_assignment2_genai/src/data.py
  inflating: /content/gpu_assignment2_genai/src/ddp_experiment.py
  inflating: /content/gpu_assignment2_genai/src/error_analysis.py
  inflating: /content/gpu_assignment2_genai/src/gpu_info.py
  inflating: /content/gpu_assignment2_genai/src/memory_experiments.py
  inflating: /content/gpu_assignment2_genai/src/serving.py
  inflating: /content/gpu_assignment2_genai/src/utils.py
  inflating: /content/gpu_assignment2_genai/src/__init__.py
/content/gpu_assignment2_genai
AI_USE.md    GPU_HOURS.md  README.md         results      src
checkpoints  HW4_5.ipynb   requirements.txt  RUN_LOG.txt  system
figures      METRICS.md    RESERVATION.md    scripts      WRITTEN.md
```

```
import torch
print("GPU:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "NONE — set Runtime to T4 G
```

```
GPU: Tesla T4
```

```
!pip install -q transformers datasets accelerate scikit-learn pandas matplotlib tqdm evaluate safetensors
!pip install -q vllm
```

```
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed.
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.34.2 which is incompatible.
cudf-cu13 26.6.0 requires numba<0.65.0,>=0.60.0, but you have numba 0.65.0 which is incompatible.
cuml-cu13 26.6.0 requires numba<0.65.0,>=0.60.0, but you have numba 0.65.0 which is incompatible.
google-adk 2.7.1 requires opentelemetry-api<=1.42.1,>=1.39, but you have opentelemetry-api 1.45.1 which is
google-adk 2.7.1 requires opentelemetry-sdk<=1.42.1,>=1.39, but you have opentelemetry-sdk 1.45.1 which is
gradio 6.30.0 requires pillow<13.0,>=8.0, but you have pillow 13.0.0 which is incompatible.
numba-cuda 0.28.2 requires cuda-core<1.0.0,>=0.5.1, but you have cuda-core 1.2.2 which is incompatible.
```

```
!python scripts/00_capture_system.py
!python scripts/01_bert_finetune.py
!python scripts/03_vllm_benchmark.py
!python scripts/04_part_a_batch_scaling.py
!python scripts/05_part_b_memory_techniques.py --batch-size 8 --epochs 2
```

```
!python scripts/07_part_d_serving_and_kv.py
```

```
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.7.self_attn
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.8.input_laye
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.8.mlp.down_p
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.8.post_atter
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.8.self_attn
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.9.input_laye
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.9.mlp.down_p
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.9.post_atter
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight layers.9.self_attn
(EngineCore pid=21673) WARNING 10-26 22:31:45 [utils.py:333] Attempted to load weight norm.weight with dt
Loading safetensors checkpoint shards: 100% 1/1 [00:01<00:00,  1.84s/it]
(EngineCore pid=21673) INFO 10-26 22:31:45 [default_loader.py:501] Loading weights took 1.87 seconds
(EngineCore pid=21673) INFO 10-26 22:31:47 [model_runner.py:424] Model loading took 0.93 GiB memory and 9
(EngineCore pid=21673) WARNING 10-26 22:31:47 [topk_topp_sampler.py:131] FlashInfer top-p/top-k sampling
(EngineCore pid=21673) INFO 10-26 22:31:47 [utils.py:347] Using LBNHC KV cache layout.
(EngineCore pid=21673) INFO 10-26 22:31:49 [monitor.py:40] [backbone] Starting torch.compile
(EngineCore pid=21673) INFO 10-26 22:31:55 [backends.py:1104] [backbone] Using cache directory: /root/.ca
(EngineCore pid=21673) INFO 10-26 22:31:55 [backends.py:1166] [backbone] Dynamo bytecode transform time:
[rank0]:W1026 22:32:01.370000 21673 torch/_inductor/utils.py:1953] Not enough SMs to use max_autotune_gem
(EngineCore pid=21673) INFO 10-26 22:32:04 [backends.py:398] [backbone] Compiling a graph for compile ran
(EngineCore pid=21673) INFO 10-26 22:32:07 [backends.py:927] collected artifacts: 25 entries, 3 artifacts
(EngineCore pid=21673) INFO 10-26 22:32:07 [decorators.py:743] [backbone] saved AOT compiled function to
(EngineCore pid=21673) INFO 10-26 22:32:07 [monitor.py:62] [backbone] torch.compile took 18.55 s in total
(EngineCore pid=21673) INFO 10-26 22:32:07 [monitor.py:91] [backbone] Initial profiling/warmup run took 6
Capturing CUDA graphs (PIECEWISE): 100% 5/5 [00:02<00:00,  1.99it/s]
Capturing CUDA graphs (FULL): 100% 2/2 [00:00<00:00, 24.93it/s]
(EngineCore pid=21673) INFO 10-26 22:32:14 [model_runner.py:1127] Graph capturing finished in 4 secs, toc
(EngineCore pid=21673) INFO 10-26 22:32:14 [gpu_worker.py:759] Available KV cache memory: 11.73 GiB
(EngineCore pid=21673) INFO 10-26 22:32:14 [gpu_worker.py:774] CUDA graph memory profiling is enabled (de
(EngineCore pid=21673) INFO 10-26 22:32:14 [kv_cache_utils.py:2327] GPU KV cache size: 1,024,800 tokens,
(EngineCore pid=21673) INFO 10-26 22:32:14 [kernel_warmup.py:176] JIT kernel warmup starting.
(EngineCore pid=21673) INFO 10-26 22:32:14 [kernel_warmup.py:189] JIT kernel warmup finished in 0.00s.
Capturing CUDA graphs (PIECEWISE): 100% 5/5 [00:00<00:00, 17.72it/s]
Capturing CUDA graphs (FULL): 100% 4/4 [00:00<00:00, 21.67it/s]
(EngineCore pid=21673) INFO 10-26 22:32:36 [model_runner.py:1127] Graph capturing finished in 2 secs, toc
(EngineCore pid=21673) INFO 10-26 22:32:36 [gpu_worker.py:971] CUDA graph pool memory: 0.03 GiB (actual),
(EngineCore pid=21673) INFO 10-26 22:32:36 [gpu_worker.py:1034] Free memory on device (14.32/14.56 GiB) c
(EngineCore pid=21673) INFO 10-26 22:32:36 [jit_monitor.py:84] Kernel JIT monitor activated; monitored JI
(EngineCore pid=21673) INFO 10-26 22:32:37 [core.py:395] init engine (profile, create kv cache, warmup mo
Downloading bytes:           |  0.00B
Reconstructing (incomplete total...): |          |  0.00B /  0.00B

Fetching 9 files: 100% 9/9 [00:00<00:00, 1472.43it/s]
Download complete:           |  0.00B
Download complete:           |  0.00B
Reconstruction complete: |          |  0.00B /  0.00B
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher r
(EngineCore pid=21673) INFO 10-26 22:32:39 [kernel.py:467] Final IR op priority after setting platform de
Rendering prompts: 100% 8/8 [00:00<00:00, 85.05it/s]
Processed prompts:   0% 0/8 [00:00<?, ?it/s, est. speed input: 0.00 toks/s, output: 0.00 toks/s](EngineCo
Processed prompts: 100% 8/8 [00:02<00:00,  2.87it/s, est. speed input: 14090.48 toks/s, output: 22.95 tok
[kv] L=8192 OK  peak=0.008 GB
(MainProcess pid=16329) INFO 10-26 22:32:42 [utils.py:628] [shutdown] Process manager: send sigterm to pr
(EngineCore pid=21673) INFO 10-26 22:32:42 [core.py:1454] [shutdown] EngineCore: trigger received signal=
(EngineCore pid=21673) INFO 10-26 22:32:42 [core.py:1635] [shutdown] EngineCore: start mode=abort timeout
(EngineCore pid=21673) INFO 10-26 22:32:42 [core.py:1666] [shutdown] EngineCore: request processing compl
(EngineCore pid=21673) INFO 10-26 22:32:42 [core.py:1499] [shutdown] EngineCore: exiting busy loop
```

```
%cd /content/gpu_assignment2_genai
!zip -r /content/colab_card2_outputs.zip results system figures
from google.colab import files
files.download("/content/colab_card2_outputs.zip")
```

```
/content/gpu_assignment2_genai
  adding: results/ (stored 0%)
  adding: results/train_partA_bs8.json (deflated 54%)
  adding: results/train_partA_bs8_epochs.csv (deflated 29%)
  adding: results/train_partB_ckpt.json (deflated 59%)
  adding: results/train_partA_bs4.json (deflated 54%)
  adding: results/train_partB_baseline.json (deflated 59%)
  adding: results/train_partB_combo.json (deflated 58%)
  adding: results/train_partB_combo_epochs.csv (deflated 36%)
  adding: results/train_partA_bs4_epochs.csv (deflated 29%)
  adding: results/kv_cache_Tesla_T4.csv (deflated 79%)
  adding: results/train_partB_accum_epochs.csv (deflated 35%)
  adding: results/train_partA_bs32_epochs.csv (deflated 28%)
  adding: results/train_partA_bs2_epochs.csv (deflated 29%)
  adding: results/train_partB_baseline_epochs.csv (deflated 35%)
  adding: results/train_partB_accum.json (deflated 58%)
  adding: results/serving_Tesla_T4.json (deflated 76%)
  adding: results/train_partA_bs1_epochs.csv (deflated 24%)
  adding: results/data_cache/ (stored 0%)
  adding: results/data_cache/split_indices.json (deflated 79%)
  adding: results/data_cache/tensors.pt (deflated 94%)
  adding: results/train_partB_8bit.json (deflated 59%)
  adding: results/serving_Tesla_T4.csv (deflated 67%)
  adding: results/train_partA_bs16.json (deflated 53%)
  adding: results/train_partA_bs128_epochs.csv (deflated 24%)
  adding: results/partA_batch_scaling_Tesla_T4.csv (deflated 58%)
  adding: results/train_partA_bs64_epochs.csv (deflated 29%)
  adding: results/hw4a_baseline_summary.json (deflated 56%)
  adding: results/slice_meta.json (deflated 41%)
  adding: results/error_gallery.md (deflated 53%)
  adding: results/train_partA_bs128.json (deflated 54%)
  adding: results/train_hw4a_baseline.json (deflated 59%)
  adding: results/partB_techniques_Tesla_T4.csv (deflated 53%)
  adding: results/train_partA_bs256.json (deflated 52%)
  adding: results/partA_batch_scaling_Tesla_T4.json (deflated 79%)
  adding: results/train_partB_amp_epochs.csv (deflated 38%)
  adding: results/train_partA_bs2.json (deflated 54%)
  adding: results/train_hw4a_baseline_epochs.csv (deflated 39%)
  adding: results/train_partA_bs64.json (deflated 54%)
  adding: results/error_gallery.json (deflated 66%)
  adding: results/train_partB_8bit_epochs.csv (deflated 38%)
  adding: results/train_partB_amp.json (deflated 58%)
  adding: results/train_partA_bs16_epochs.csv (deflated 25%)
  adding: results/partB_ranking_Tesla_T4.json (deflated 64%)
  adding: results/train_partB_ckpt_epochs.csv (deflated 36%)
  adding: results/train_partA_bs32.json (deflated 54%)
  adding: results/train_partA_bs1.json (deflated 54%)
  adding: results/kv_cache_Tesla_T4_summary.json (deflated 16%)
  adding: system/ (stored 0%)
  adding: system/gpu_snapshot_Tesla_T4_bfa082ed-9aa.json (deflated 41%)
  adding: system/gpu_basic_Tesla_T4_bfa082ed-9aa.txt (deflated 20%)
  adding: system/nvidia_smi_q_Tesla_T4_bfa082ed-9aa.txt (deflated 79%)
  adding: figures/ (stored 0%)
  adding: figures/confusion_matrix.png (deflated 14%)
  adding: figures/hw4a_training_curves.png (deflated 8%)
  adding: figures/partA_throughput_Tesla_T4.png (deflated 11%)
```

```
Start coding or generate with AI.
```